THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET C

Chapters 1 & 2: **Real Numbers, Polynomials** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The LCM of 6, 8 and 12 is: 1

(a) 12 **(b)** 24 **(c)** 48 **(d)** 96

2. The HCF of $2^4 \times 3$ and $2^2 \times 3^2 \times 5$ is: 1

(a) 4 **(b)** 720 **(c)** 12 **(d)** 6

3. If $a = 2^2 \times 3$ and $b = 2 \times 3^2$, then LCM(a, b) is: 1

(a) 18 **(b)** 36 **(c)** 72 **(d)** 6

4. A student states: "$\frac{1}{\sqrt{2}}$ is rational because it is written as a fraction." The correct response is: 1

(a) Incorrect — $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ is irrational **(b)** Correct

(c) Correct for even numbers **(d)** Cannot be determined

5. Student A says HCF $\times$ LCM $=$ product holds for any two positive integers; Student B says it holds for any three. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The zeroes of $x^2 - 1$ are: 1

(a) no real zeroes **(b)** 1 only **(c)** ± 1 **(d)** 0, 1

7. If 2 is a zero of $x^2 + ax - 8$, then a is: 1

(a) −2 **(b)** −4 **(c)** 2 **(d)** 4

8. A quadratic polynomial with sum of zeroes $-\frac{1}{4}$ and product $\frac{1}{4}$ is: 1

(a) $4x^2 + x - 1$ **(b)** $4x^2 + x + 1$ **(c)** $x^2 + 4x + 1$ **(d)** $4x^2 - x + 1$

9. A parabola $y = p(x)$ meets the x-axis at $(-2, 0)$ and $(5, 0)$. The zeroes of $p(x)$ are: 1

(a) $2, -5$ **(b)** $-2, 0$ **(c)** $-2, 5$ **(d)** $0, 5$

10. A student states: "The zeroes of $x^2 + 4$ are ± 2." The correct response is: 1

(a) Only 2 is a zero

(b) Correct

(c) Only −2 is a zero

(d) Incorrect — $x^2 + 4 > 0$, so it has no real zero

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the HCF of 867 and 255 using prime factorisation. 2

12. Classify as rational or irrational with reason: **2**

(i) $(3+\sqrt{7})-\sqrt{7}$

(ii) $(\sqrt{2}+\sqrt{3})^2$.

13. Find the zeroes of $x^2-(\sqrt{3}+1)x+\sqrt{3}$. **2**

14. If α, β are the zeroes of x^2+3x+1, find $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{13}$ is irrational. **3**

16. If the zeroes of x^2+px+q are double the zeroes of x^2-5x+6, find p and q. **3**

17. The LCM of two numbers is 2079 and their HCF is 27. If one number is 189, find the other. **3**

OR

Find the smallest number which leaves remainder 3 when divided by 5, 6 and 8. **3**

18. Find the zeroes of $4\sqrt{3}x^2+5x-2\sqrt{3}$ and verify the relationship between the zeroes and the coefficients. **3**

OR

If 1 and -2 are two zeroes of x^3+ax^2+bx-2, find a, b and the third zero. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** From a bus stand, buses to Ajmer leave every 30 minutes, to Kota every 45 minutes and to Udaipur every 60 minutes. All three leave together at 6:00 a.m. **5**

(i) Write the prime factorisation of 30, 45 and 60.

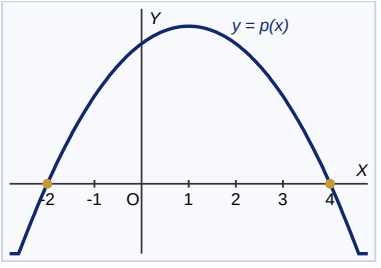
(ii) When will they next leave together?

(iii) How many times will they leave together from 6:00 a.m. to 9:00 p.m. (both included)?

OR

Prove that $\sqrt{3}$ is irrational. Hence show that $5+2\sqrt{3}$ is irrational. **5**

20. **Case study:** The graph of $y=p(x)$ shown is a parabola. **5**



(i) Write the zeroes of $p(x)$.

(ii) If $p(x)=-x^2+bx+c$, find b and c.

(iii) Verify the relationship between zeroes and coefficients.

(iv) Find the maximum value $p(1)$.

OR

If α, β are the zeroes of $2x^2-5x+2$, find **5**

(i) $\alpha^2+\beta^2$

(ii) $(\alpha-\beta)^2$

(iii) $\frac{1}{\alpha}+\frac{1}{\beta}$.

(iv) Write a polynomial whose zeroes are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

END OF QUESTION PAPER — **Total: 40 Marks**